Shubham Trivedi

Senior Embedded Engineer | Embedded Linux & Yocto

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PROFILE

Embedded Linux and Yocto engineer with 7+ years of experience developing Linux-based systems on ARM platforms. Experienced in Yocto/OpenEmbedded, BSP and kernel customization, Linux user-space applications, system integration, secure software updates, and CI/CD automation. Work spans C, C++, Python, and shell scripting, from hardware interfaces to build and deployment tools.

TECHNICAL SKILLS

Linux & BSP	Yocto Scarthgap, OpenEmbedded, BitBake, kas, U-Boot, kernel configuration, DTS/DTB, RootFS customization
Development	C, C++, Python, Bash, Qt5, CMake, systemd, IPC, gRPC, Protocol Buffers
Build & testing	Azure DevOps, GitHub Actions, Docker, QEMU, kas-container, sstate/download mirrors, PyTest, Google Test, Robot Framework
Updates & security	SWUpdate, read-only RootFS, Linux permissions, DAC, JWT
Platforms & protocols	ARM, AM64xx, i.MX8, UART, SPI, I2C, BLE/BlueZ, TCP/IP

PROFESSIONAL EXPERIENCE

Atlas Copco
March 2024 – Present

Senior Engineer, Embedded Systems

- Develop custom Yocto Scarthgap distributions, layers, and recipes for AM64xx and i.MX8 using kas.
- Build C++/Qt5 applications and port Windows applications to Linux using CMake and Yocto devtool.
- Implement SWUpdate and read-only root filesystem solutions; develop Python tools for flashing, deployment, and validation.
- Maintain Azure DevOps build/test/release pipelines and automated tests with PyTest, Google Test, and Robot Framework.

LG Soft India
August 2023 – March 2024

Senior Research Engineer

- Developed C++ Linux applications for hardware API virtualization and production validation on LG webOS.
- Used Yocto for firmware builds and software integration; collaborated on system-level debugging.

Apexon
March – July 2023

Senior Engineer

- Developed Linux applications with C++, gRPC, Protocol Buffers, and JWT for secure communication and backend integration.
- Implemented automated tests using Google Test.

Inpixon
January 2020 – March 2023

Embedded System Engineer

- Developed ARM/Yocto firmware, SWUpdate upgrades, and boot optimizations; updated kernel configurations and DTS files for legacy hardware.
- Built C/C++ sensor and TCP applications, plus microcontroller upgrade and hardware validation tools.
- Implemented GitHub Actions CI automation and debugged firmware and hardware/software integration.

Wideeye Technologies April – December 2019
IoT Embedded Software Developer

- Developed Python/BlueZ BLE data collection applications on Linux.
- Optimized embedded algorithms, responsiveness, and power use in battery-operated systems.

INTERNSHIP

Transpipe Integrity Solutions July – November 2018
Embedded System Trainee

Contributed to an embedded pipeline-monitoring system and developed low-level drivers for peripherals including IMU and NAND flash.

PERSONAL PROJECT

yocto-board-lab — In development

Developing a reusable Yocto BSP framework for ARM and RISC-V SBCs with reproducible kas/kas-container builds. Areas of focus include QEMU validation, PREEMPT_RT experimentation, AI/NPU-capable boards, CI/CD integration, and documentation.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering
Sri Sairam College of Engineering · 2019

Diploma, Electronics and Telecommunication Engineering
Government Polytechnic Jalgaon · 2014

COURSES & CERTIFICATIONS

- Embedded Linux Device Driver Developer – Level 2, Kernel Masters · November 2021 – February 2022
- Embedded Linux Device Driver Developer – Level 1, Kernel Masters · September – November 2021
- Additional coursework: A Beginners Guide to Django; Red Hat Enterprise Linux Technical Overview

RECOGNITION

- Smart India Hackathon 2018 winning team, Mumbai.
- Inpixon Rock-Star of the Month for resolving a complicated firmware problem.
- Organizer and resource person for an embedded systems workshop.

LANGUAGES

Marathi, Hindi, English